Short Questions

Course Name: Professional AI Engineering

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1. AI mainly creates new content such as text, images, code, or audio based on user input. For example, ChatGPT can generate answers, write emails, or create articles. It focuses on producing content from the given prompt.

On the other hand, Agentic AI can make decisions and perform actions automatically to complete a task. It does not only generate responses but can also interact with tools, databases, or applications.

A real-world business example is customer support service. A Generative AI chatbot can reply to customer questions by generating text answers. However, an Agentic AI system can check order details, process refunds, send emails, and solve customer problems automatically without much human help.

So, the main difference is that Generative AI creates content, while Agentic AI can take actions and complete tasks intelligently.

2. CPU and GPU both are important for local AI development, but their performance is different. A CPU is designed for general tasks and usually works with a smaller number of cores. It can run AI models, but the speed is slower, especially for large models and heavy processing.

On the other hand, a GPU has many cores and can process multiple tasks at the same time. Because of this, GPUs are much faster for AI training and running large language models. Local AI projects usually perform better with a GPU because it reduces processing time and gives faster results.

Docker is also important in the development workflow. It helps developers create a separate and consistent environment for the project. With Docker, the same application can run easily on different computers without software conflicts. It also makes installation, dependency management, and project sharing easier for developers and teams.

3. Quantization Methods are techniques used to reduce the size of Large Language Models (LLMs). In this process, large data formats like 32-bit numbers are converted into smaller formats such as 8-bit or 4-bit. This helps make the model lighter and easier to run on local computers.

Quantization is very useful for running local LLMs because large models normally require a lot of RAM and GPU memory. After quantization, the model uses less memory, so it can run even on devices with limited hardware resources.

It also helps in latency optimization. Since the model becomes smaller, it can process data faster and generate responses more quickly. As a result, users get faster output and smoother performance while using local AI applications.

4.Embeddings are a method of converting text into numerical form (vectors) so that the AI can understand the meaning of the text. Instead of looking at words directly, the model represents them as numbers that capture their semantic meaning. This helps the AI understand which texts are similar or related to each other.

Retrieval Methods are used to find the most relevant information from a database or document collection based on a user's question. The system compares the question's embedding with stored embeddings and selects the closest matches.

In a local AI project, embeddings and retrieval work together. Embeddings help represent the meaning of data, and retrieval methods use that representation to quickly search and fetch the most relevant information. This combination is very important in systems like RAG, where accurate and context-based answers are needed.

5. RAG stands for Retrieval-Augmented Generation. It is an AI system that combines information retrieval with text generation. In this architecture, the system first searches for relevant information from documents or a database, and then the AI model uses that information to generate an answer.

The process starts when a user asks a question. After that, the system retrieves related data from stored documents. Finally, the language model creates a response based on the retrieved information. This helps the AI provide more accurate and updated answers.

Document Processing is an important part of the RAG system. It prepares documents by cleaning, formatting, and organizing the data so the AI can understand it properly.

Chunking Strategy means dividing large documents into smaller parts or sections. This makes searching easier and helps the system find the most relevant information quickly. As a result, the generated answers become more accurate and efficient.